

Overview :

During this research project, I worked on two completely different domain-specific language models: MediGPT for the medical domain and DungeonMasterGPT for fantasy-style storytelling and roleplay generation. The main objective behind working on two separate domains was not only to implement GPT-style transformers from scratch, but also to understand how transformer-based language models behave when trained on highly specialized datasets with very different linguistic structures, vocabulary distributions, and contextual reasoning patterns.

The project was heavily inspired by the “Attention Is All You Need” paper and the Vizvara “Building LLMs from Scratch” course, where the focus was on understanding the internals of decoder-only transformers instead of simply using pretrained APIs. Through these implementations, I was able to compare how transformer attention mechanisms adapt differently in technical biomedical text versus imaginative fantasy narratives.

For the medical domain, I developed MediGPT, a small GPT-style transformer trained on medical corpora such as MedQuAD and PubMedQA. The goal was to study whether a relatively small language model could learn clinical terminology, disease relationships, and structured medical reasoning patterns. One of the biggest challenges in this domain was preprocessing. Standard cleaning methods often removed important biomedical terms such as “covid-19”, “anti-inflammatory”, or dosage expressions. To solve this, I implemented a medical-aware preprocessing pipeline that preserved clinically meaningful text while removing noise such as HTML tags, URLs, and unwanted symbols.

The final MediGPT training corpus consisted of approximately 15,000 medical documents and nearly 6.5 million tokens. GPT-2 BPE tokenization using tiktoken was used to handle rare medical vocabulary through subword decomposition. During training, the model gradually learned meaningful relationships between symptoms, treatments, diseases, and risk factors. Generated outputs demonstrated understanding of concepts such as diabetes complications, hypertension management, and cardiovascular risk factors.

In parallel, I also worked on DungeonMasterGPT, which focused on fantasy storytelling, RPG-style narratives, magical world-building, and dialogue generation. Unlike the structured and factual nature of medical text, the fantasy dataset required learning creativity, narrative continuity, character interaction, and descriptive language. This domain highlighted a different side of transformer behavior, where coherence, imagination, and contextual continuity became more important than factual correctness.

Working on both projects gave me practical insight into how domain specialization affects token distributions, attention patterns, generation quality, and training stability. It also showed that transformer architectures are highly adaptable and can learn completely different language behaviors depending on the training corpus.

The implementation files for the project include:

- MediGPT_Research_Paper_v2.tex — Main medical research paper
- MediGPT_Research_Discussion_Section.tex — Medical Q&A discussion section
- DungeonMasterGPT_Research_Paper.tex — Main fantasy-domain research paper
- DungeonMasterGPT_Research_Discussion_Section.tex — Fantasy Q&A discussion section
- DungeonMasterGPT_Research_Notebook (2).ipynb — Complete transformer implementation and training notebook
- Sample Outputs PDF — Generated text examples from both models

Overall, this project helped me understand transformer architectures at a much deeper level by implementing, training, debugging, and analyzing domain-specific language models entirely from scratch.